DEVELOPMENT WORK REPORT

SmartAg Ops Advisor

Every request made during the development session, what was built or fixed, and how it was verified.

Application	SmartAg Ops Advisor (FastAPI backend · React frontend · PostgreSQL on Neon)
Session	24–25 October 2026
Requests covered	23
Repository	`Data-Drive/agri-project`
Companion documents	`SECURITY.md`, `docs/MASTER_DOCUMENT.md`

Requests are paraphrased from the original prompts; pasted HTML snippets are summarised. API keys, tokens and passwords that appeared in prompts are deliberately not reproduced in this report.

1. Summary

The session started from a folder whose source code had been deleted. The application was recovered from git, brought up on a hosted PostgreSQL database, and then extended request by request: admin two-step sign-in, driver document uploads, an admin dashboard with usage charts, voice input in ten languages, a security policy, layout fixes, government mandi price data, and a market finder that recommends where to sell after real road distance and costs.

Area	Outcome
Recovery & running	Source restored from git objects; backend, frontend and database running; data on Neon
Accounts & admin	Admin email OTP sign-in, user removal, admin-only menu, admin dashboard with charts
Truck operators	Permit + driving licence (front/back) uploads stored in the database, inside a collapsible Edit profile form
Maps	Google-Maps-style full screen on Live Tracking and Find Markets (map/satellite, re-centre)
AI assistant	Voice input (10 languages); chat no longer hangs (automatic fallback model)
Market data	Government Agmarknet prices imported from the uploaded dataset; market finder on real roads and costs
Quality & security	Security policy enforced and documented; alignment fixed across all pages; master document written

2. Requests at a glance

#	Request	Status
1	Install the Superpowers plugin from the provided install instructions	**Done**
2	Run the application frontend, backend and database	**Done**
3	Add the Gemini API key	**Done**
4	Switch to the Neon database and remove the local one	**Done**
5	Admin login with emailed OTP (not for farmers / truck operators)	**Done**
6	Small trip map with a Google-Maps-style full-screen button	**Done**
7	Permit and driving-licence (front/back) image uploads stored in the database	**Done**
8	Merge the document uploads into the profile form; Save / Reset	**Done**
9	Edit profile opens on click; Save / Reset close it	**Done**
10	Remove My Crops, Find Markets, Truck Booking from the admin role	**Done**
11	Remove Market Prices, Live Tracking, My Bookings from the admin role	**Done**
12	Admin can remove a user (button + database)	**Done**
13	Admin dashboard: user numbers, AI chatbot usage graph, request monitor graph	**Done**

14	Microphone voice recognition with multi-language support and a proper backend	**Done**
15	Add the mandatory application security policy	**Done**
16	Fix the application alignment fully	**Done**
17	Master document for the application	**Done**
18	Daily government mandi prices from data.gov.in (original data only)	Built; superseded by #19
19	Use the uploaded dataset for mandi prices	**Done**
20	Find Markets: best market after transport and labour, with map and backend	**Done**
21	Full-screen button on the Find Markets map	**Done**
22	AI assistant chat stuck loading	**Fixed**
23	Convert the full set of prompts into a PDF report	**This document**

3. Requests in detail

1 Install the Superpowers plugin

> *"Fetch and follow instructions from the Superpowers INSTALL.md." Then chose option 2.*

WHAT WAS DONE

- The given INSTALL.md link returned 404; the current instructions were taken from the project README.
- Added the `obra/superpowers-marketplace` marketplace and installed `superpowers@superpowers-marketplace` (user scope) using the Claude CLI bundled with the VS Code extension.

2 Run the frontend, backend and database

> *"Run the application frontend and backend and database."*

WHAT WAS FOUND

The project folder contained no source code — only a broken Python virtual environment, a Vite cache, logs and a `.git/objects` folder. Logs showed the files had disappeared that afternoon.

WHAT WAS DONE

- Rebuilt the repository from the 604 git objects and restored the latest commit (95 files: FastAPI backend + React frontend), without overwriting anything.
- Recreated the Python virtual environment, installed npm packages, created `.env` files from the templates with a new random secret key.
- Created a local PostgreSQL database and started both servers (API on port 8000, app on port 5173).

Edits made after the last commit (14:03 that day) and the original `.env` files could not be recovered.

3 Gemini API key

> *Provided a Gemini API key.*

- Stored in the backend `.env` (removing a stray space), verified with a live Gemini request, backend restarted.

4 Neon database

> *Provided a Neon PostgreSQL connection string: "remove the old local database, add the new database."*

- Connection tested first: Neon already held the project's tables and users.
- `DATABASE_URL` switched to Neon; the local `agri` database dropped.
- Added connection health checks (`pool_pre_ping` , recycle) to stop the "SSL connection closed unexpectedly" errors seen in the old logs.

5 Admin sign-in with emailed OTP

> *"Admin logs in on the normal login page with the admin email and default password; after Login an OTP is sent. Not for farmers and truck operators." (Mailtrap sending example provided.)*

- Admin account created automatically from configuration; after a correct password, a 6-digit code is emailed via Mailtrap (10-minute expiry, 5 attempts, 60-second resend).
- New OTP step on the login page; farmers and truck operators sign in directly.
- Fixed two security gaps found while testing: the temporary sign-in token worked as a full session, and public sign-up accepted the admin role.
- The provided sender domain was not verified in Mailtrap, so the working demo sender was kept.

Verified: live email sent; wrong code, reused code, early resend and non-admin logins all behave correctly.

6 Trip map: small view with Google-Maps-style full screen

> *"The map should be small, with a button to go full screen like Google Maps."*

- Live Tracking map reduced in size, with Google-style zoom buttons and a full-screen button.
- Full screen: floating trip card, round close button, re-centre, Map/Satellite switch; Esc to exit; fallback for iPhone Safari.

7 Driver documents stored in the database

> *"Add permit and driving licence uploads; the licence has two sides; images stored in the database."*

- New `user_documents` table and API: vehicle permit, licence front, licence back (JPG/PNG/WebP, 5 MB, real image check).
- Upload cards with previews on the truck operator's profile; admins can view documents through the API.

8 Documents inside the profile form

> *"The fields should be merged in one box; Save will save and Reset will reset."*

- Document slots moved into the Edit profile form; picked files are staged ("Unsaved") and uploaded or removed only on **Save changes**; **Reset** discards them.

9 Collapsible Edit profile

> *"The user clicks Edit profile to open it; otherwise it stays closed."*

- Edit profile starts closed; clicking the header opens it; Save (on success) and Reset close it.

10–11 Admin menu limited to admin work

> *"Remove My Crops, Find Markets, Truck Booking" and then "Market Prices, Live Tracking, My Bookings" — for the admin role only.*

- Hidden from the admin menu and blocked by route guards; all other roles unchanged.
- Admin menu: Dashboard, Reports, AI Assistant, Profile, Admin.

12 Admin can remove users

> *"Admin can remove a user — remove it in the database too, with a button."*

- Remove button with inline confirmation on the Admin page; backend deletes the user and their crops, bookings, notifications, chats and documents (database cascades verified on Neon).
- Safeguards: cannot remove yourself or the last admin; the removed operator's trucks become inactive; old sessions stop working.

13 Admin dashboard

> *"Show how many users, farmers and truck operators; an AI chatbox usage graph; an application request monitor graph."*

- New admin dashboard: total users (hero number), farmers, truck operators, AI chat users; users-by-role bars.
- AI chatbot usage chart (questions over time) and request monitor chart (requests and server errors, average response time, 4xx/5xx).
- Request metrics are counted in memory and saved hourly to the database every minute; charts have hover tooltips, keyboard support and a table view; 24 h / 7 d / 30 d filter.

14 Voice input in the AI assistant

> *"Add the mic — voice recognition with multiple language support and a proper backend."*

- Mic button records up to 60 s, converts to WAV in the browser, and the backend transcribes it with Gemini in the selected app language (10 Indian languages), without translating.
- Uses the faster lite model with automatic fallback; text is placed in the message box for review before sending.

15 Application security policy

> *"Add the application security policy across the full application — whatever is mandatory."*

- Password policy, account lockout (5 failures → 15 minutes), rate limits on sign-in/OTP/AI endpoints, session revocation after password reset.
- Security headers and Content-Security-Policy on API and frontend, strict CORS, request size limit, production mode (HSTS, no API docs, refuses weak secrets).
- Security audit log; written policy in `SECURITY.md` with a deployment checklist.

16 Alignment across the application

> *"Fix the application alignment fully."*

- Every page checked for all three roles at desktop and phone width with screenshots and automatic overflow / stretched / clipped-element checks.
- Fixed: 130 px-tall filter inputs on phones, misaligned Refresh buttons, empty hero space, clipped booking cards, wrapping tabs, truncated language selector, a missing table header, cramped stat cards.

17 Master document

> *"Give me a master document for my application."*

- `docs/MASTER_DOCUMENT.md`: overview, roles, features, journeys, architecture, data model, business rules, full API reference, AI, security, configuration, setup, deployment, operations, limitations and roadmap.

18 Daily government mandi prices from data.gov.in

> *"Use the data.gov.in mandi price resource with my API key, updated every day; original data only, no fake data."*

- Built a sync that downloads the full daily feed (≈15,000 records, 27 states) into a `mandi_prices` table, all-or-nothing, with retries for the API's outages and an API-key redaction fix.
- Stopped at the user's request in favour of the uploaded dataset (#19); the API mode remains available via a setting.

19 Use the uploaded dataset

> *"I uploaded the data in the dataset folder — use it."*

- CSV files in `dataset/` are imported exactly as published (173 Tamil Nadu tomato prices) and re-imported automatically when a file is added or changed.
- New "Government mandi prices" tab on Market Prices with source, date, filters, min/max/modal ₹ per quintal and ₹/kg.

Verified: all 173 stored prices match the CSV value for value.

20 Find Markets — best market for the farmer

> *"Backend calculates the best nearby market after comfortable transport charge and labour; list Nearest, Quickest trip, Best crop price, best profit; Leaflet map with a proper backend."*

- Market locations from OpenStreetMap (cached; district centre used and labelled when a market isn't found); real road distance and drive time from OSRM.
- Transport priced with the cheapest truck listed in the app (base fare + ₹/km); labour, commission, packaging and loss from configured rates.
- Four picks (Best profit, Best crop price, Nearest, Quickest), ranked list, cost breakdown, Leaflet map with road route; pickup from GPS, typed place, map tap or crop pickup location.

Example: from Trichy, 500 kg tomato — Perambalur (58 km, ₹5,000/q) leaves ₹21,555 in hand, versus ₹16,630 at the nearest market.

21 Full screen on the Find Markets map

> *"Add the full-screen button like Google Maps."*

- Same Google-Maps-style full screen as Live Tracking, with a selected-market card (money in hand, distance, time, price), route re-centre and Map/Satellite switch.

22 AI assistant stuck loading

> *"The AI assistant chatbox is not working properly — the chat keeps loading."*

- Cause: the main Gemini model was overloaded and the app waited up to 3 minutes before failing.
- Fix: switch to the faster model automatically if the main one hasn't started answering within 20 s; the chat page shows "Still thinking…" after 12 s and stops waiting after 75 s of silence.

Verified: answers now arrive in about 10–30 s in English and Tamil, including through the real chat page.

23 This report

> *"Convert my full prompt to generate the PDF report."*

- All requests of the session summarised with outcomes and verification, generated as this PDF.

4. Open items and recommendations

Item	Recommendation
Secrets shared in chat	Rotate the Neon database password, Gemini key, Mailtrap token and data.gov.in key; change the default admin password (it does not meet the new password policy).
Version control	None of the session's changes are committed yet — review and commit them.
Automated tests	The repository has no automated test suite; add API tests (pytest) and key journey tests (Playwright).
Market data coverage	The uploaded dataset covers only tomato in Tamil Nadu; add more CSV files or switch the source to the data.gov.in API for all crops and states.
Free map services	OpenStreetMap geocoding and the public OSRM server are suitable for development and modest use; use a paid or self-hosted routing service for production load.
Single server process	Rate limits and request metrics are kept per process; run one backend worker or move them to a shared store (e.g. Redis) before scaling out.
Backend auto-reload	The development server's auto-reload hung several times on Windows and needed restarts; production should run without `--reload`.
Translations	New interface text is in English; regenerate the language files to translate it.

Generated from the development session of 24–25 October 2026.